



YOUR KINDLE NOTES FOR:

Crux (The Nexus Trilogy Book 2)

by Ramez Naam

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84 Highlights | 4 Notes

Highlight (Yellow) | Location 175

You can't do this! What gives you the right? I made this, Kade told the man. That gives me the right.

Highlight (Yellow) | Location 374

Ling could see the lights of the city out that window, now, the twenty-story-tall female face on the skyscraper across the street, winking and smiling, advertising some product for the humans to consume. The world inside her felt more real.

Highlight (Yellow) | Location 379

A new bounty on her friend Kade's head, announced by the Americans.

Highlight (Yellow) | Location 457

Then she began to meditate. She started with anapana, the meditation of the breath, then worked to vipassana, meditation of awareness of the body. Her mind stilled, she turned at last to the three thousand year-old practice called metta, the meditation of loving-kindness.

Highlight (Yellow) | Location 551

Software beings, all of them. Digital representation of brains. Like her. What mattered was pattern, not substrate. A physical brain was an information processor and nothing more. A mind was the information being processed, not the physical brain that did the processing. A digital brain, with digital neurons and digital synapses and digital signals passing through it, could process that information just the same, could give rise to a mind just as well. Provided, of course, that the underlying model of neurons and synapses and all the rest of the brain was accurate.

Highlight (Yellow) | Location 563

Her imminent body-death had forced Chen and Thanom to try the one thing that might save her mind: uploading her, using the technology they three had been building. The perfect team – Thanom Prat-Nung, the Thai nano-

engineer with his molecular devices that could scan a brain at nanometer scale; her brilliant husband Chen with his quantum computing cluster powerful enough to simulate a human brain; and her, the neuroscientist with the mathematical model to run that uploaded brain.

Highlight (Yellow) | Location 571

And miracle of miracles, it worked. Her burned, broken, ruined body died, but the pattern of her brain, the precise wiring of her hundred billion neurons and the hundred trillion synaptic connections between them, was captured, simulated, and run. She awoke as software running on the massive cluster beneath Jiao Tong University. She was angry, grieving, but alive. More alive and more aware than ever.

Highlight (Yellow) | Location 694

techniques for stilling and guiding their own minds through meditation, for sinking into that egoless state where their minds, bridged by Nexus, could become one.

Highlight (Yellow) | Location 697

He'd seen beautiful things in those months, in Cambodia and across the net. The mentally and emotionally scarred, healing. Patients in comas being touched and restored to consciousness. Scientists tapping into each other's perspectives, making breakthroughs they never could have alone. Artists creating new forms that they didn't even have names for, that immersed you in experiences unlike any other.

Highlight (Yellow) | Location 700

And union. Minds coming together. Walls dropping. Consciousness spanning bodies. Group minds, self-assembled, voluntary, greater than the sum of their parts...

Highlight (Yellow) | Location 710

"Hey, Kade," Feng called from up ahead. "What did Confucius say about man who runs in front of car?" Kade smiled and shook his head, brushing away more foliage from his face. "I don't know, Feng. What?" "He gets tired!" Feng roared. "Get it? Tired?"

Highlight (Yellow) | Location 725

"Hey Kade," he called from up ahead. "You know what Confucius said about the man who runs behind a car?" Kade laughed, struggling to keep up. "What, Feng?" "He gets exhausted!" Feng sang out. "Exhausted!"

Highlight (Yellow) | Location 731

The monks greeted them as heroes, Kade as a holy man. He did his best to deflect their adoration, laugh with them, diffuse the power imbalance as always. I'm just like you, he tried to show them. Just another novice.

Highlight (Yellow) | Location 739

It was the logical direction of human evolution. Humanity had achieved what it had not through strength or claws or armor, not even through individual human intelligence, as impressive as that was. No, it was the ability of humans to coordinate, to work together, to produce ideas and solutions collectively that no individual mind ever could, that truly set them apart. Nexus was just one more step in that direction. And for the monks, it was more than that. In their view, Nexus was a spiritual tool. It helped tear down the illusion of separateness. It helped pierce the veil of maya. It helped these monks, all part of the same conscious universe, forget the lie that they were separate, the broken distinction of one person ending before the next began. By linking their minds it helped them remember that they were, in fact, all one.

Highlight (Yellow) | Location 773

Every one of these plants would be illegal in Europe, she thought. Most of them illegal back in the US. How strange to live in a place where this technology was so normal, so essential, even. Rich countries had the luxury to ban biotechnologies. Poor countries depended on them.

Highlight (Yellow) | Location 790

Eight of the children, ranging in age from one to eight years old, had been exposed to Nexus in the womb, most of them repeatedly. Once a mother felt her unborn child's mind through Nexus, most felt a strong draw to take Nexus again, to touch the half-formed thoughts of the little being growing inside themselves once more. The children were enchanting, vexing, confounding. Most of them were scarred in some way. They acted out at times, testing her, bickering with each other, being petulant or disobedient or just stubborn. But they were also radiant at a level that shone right through their scars and the trouble they caused. Their use of Nexus was instinctive, fluent in a way that Sam would never be. They communicated with each other more in thoughts than words, in blurs of impressions and ideas often too fast for her to follow. And she could hide nothing from them. They knew her inside and out. The touch of their minds made her spirit soar. She couldn't get enough of them.

Highlight (Yellow) | Location 821

his mind was the most wonderfully unique of any of them, all bright colors and moving shapes and form without meaning. The universe shimmered when she saw it through his eyes. Zen mind. Beginner's mind.

Highlight (Yellow) | Location 1187

The house was so empty now, with their sons off at college, an ocean away in Germany and France. Why had he ever let them go? To sclerotic, stagnant, backwards-looking Europe of all places? They should have gone to Asia if anywhere, to a place that looked towards the future instead of fetishizing the past.

Highlight (Yellow) | Location 1328

Instead, he told them everything. How the back doors worked, all three of them. The hacks in the compiler to bake them into Nexus even though they didn't exist in the source code. The obfuscation tricks that hid the back

doors and passwords as random ones and zeros scattered among billions in the binaries, indistinguishable from the parameters that governed millions of neurons, nearly impossible to reverse-engineer. And of course, the passwords themselves. He told them how they could build a new compiler from scratch to evade the back doors, but he could tell they didn't care. All they wanted was the current passwords, and how to use them. Which meant that they wanted to break into minds already running Nexus 5. Which meant that it had gotten out.

Highlight (Yellow) | Location 1411

He closed his eyes once more now, and let all that data pour through him. The software in his mind visualized it, placed those minds on a shadowy globe as points of light, the shape of continents visible in the diffusion of minds over the earth's surface, a pattern like nothing so much as the night-time lights of civilization as seen from space.

Highlight (Yellow) | Location 1502

It's about the money, Nakamura had taught her. Rich countries don't mind Copenhagen. But for countries still wrestling with poverty, the technology can make a huge economic difference. The motivation's higher.

Highlight (Yellow) | Location 1531

"It's like," Jake started, haltingly. "The big picture, OK? It's like the next step in our evolution. Going from one mind, to many minds, all linked... Group consciousness. That we're just parts of. That's the real post-human."

Highlight (Yellow) | Location 1534

"There was this woman," Jake said, "this PhD in the states, who used to write about this all the time. Her dissertation was about how we're already group minds, how we've been evolving in that direction already – from animal cries to language to writing to the internet, and now Nexus. How it's collective thinking makes us special. Ilyana Alexander."

Highlight (Yellow) | Location 1545

awake, and opened herself, and felt the minds of the children, dreaming and breathing in unison. So very human, to her. And maybe more. Nine minds that dreamt as one... She'd felt something like that, in Bangkok, in that loft, when she'd felt like she was part of the Buddha. She'd felt it that night, when she and Kade had fallen asleep together, had dreamt each other's dreams. She'd felt it with Ananda's monks, meditating as one mind. But all of those had been temporary, fragile states. These children... they did it naturally, automatically. Was it really possible to become part of something larger? For minds to meld together for more than just fleeting moments?

Highlight (Yellow) | Location 1843

The virus had been his idea. With the thawing Arctic belching ever more carbon into the skies, with solar shields caught up in endless political debate, with every attempt to reduce the warming and acidification of the planet

falling short of what was necessary, someone had to take the initiative. And so he had. Survival genes, culled from the coral species with the greatest tolerance to heat and acid, inserted into a viral vector that would spread those genes to every coral reef in the world, give them precious decades of additional life. And it was working, bit by bit, restoring life to nearly dead reefs, strengthening others. Just a buffer, just ten or twenty years' worth, perhaps. But at least he'd done something. And how they hated him for it. Unilateral, they called it. A crime against nature. Uncontrolled experimentation. Unsanctioned madness.

Highlight (Yellow) | Location 1900

"Cluster one: Mid-teens to mid-twenties of age, urban and suburban, medium to high income, roughly even gender split. "Recreational users," Elizabeth Broadmoor piped in. "Party kids."

Highlight (Yellow) | Location 1904

"Cluster two: Thirties to fifties, suburban, tilted towards high income, sixty per cent women. These are parents of special needs children. Autism spectrum, ADHD, etc..."

Highlight (Yellow) | Location 1906

"This is the one you asked us to look for. Highly educated, mostly high socio-economic status, fifty-five per cent male, urban and suburban. They're highly international. What unites them most are the features you suggested: High scientific literacy, high IQ, careers in engineering, computing, and the sciences. These are people seeking out Nexus for themselves, to connect and enhance their own performance. These are the intelligentsia you thought we'd find."

Highlight (Yellow) | Location 1928

Their linked brains were playing a game, or so they thought. A molecular design game, searching through genetic sequences that would yield a protein that would go even further in restoring the world's corals, in protecting them from the acidification of the seas. Shiva closed his eyes and he could see the shifting protein shapes in the children's minds, writhing, folding, refolding, transforming as the youngsters searched the possibility space for a new way to save the world's reefs. The expertise in this game came entirely from the adults – molecular biologist and biochemists with deep knowledge in the calcifying proteins used by corals. But the raw skill in the game, that came from the children, who tapped into that knowledge and then applied it together at staggering speed.

Highlight (Yellow) | Location 1941

Humanity was failing. It could not solve the problems it now faced. But those millions of Nexus-augmented minds could. They could become a single post-human intelligence of epic scale. A god forged out of humanity, finally able to manage the planet through its Anthropocene calamities. But those millions would not merge willingly or easily. Shiva would need to forge that god out of its component pieces, would need to give it direction, to turn it into the rightful governor of this world and the people on it. And for that he needed Kaden Lane.

Highlight (Yellow) | Location 1989

“Bo Tat,” he heard. “Bo Tat.” A hundred voices said it. He didn’t know the words but he could see the meaning of it in their minds, their hundred minds merged as one. Bodhisattva. Heroic-minded one. Bringer of light. He who would sacrifice himself, be reborn in suffering, time after time, until every living being reaches enlightenment.

Highlight (Yellow) | Location 1998

“I am not Bo Tat,” he told them with a laugh. Not enlightened. Not heroic. “I’m a novice. Less than a novice.”

Highlight (Yellow) | Location 2084

Feng moved like a dancer, calm and graceful. He leapt over the plane of fire of a swinging pistol, rolled under another as he converged on the first bounty hunter. His mind was utterly absorbed. This was samadhi. This was meditation. Guns exploded and the bullets were living things in Feng’s mental map, ripping out of the muzzles, shockwaves rippling visibly through the air, flinging themselves at the spots Feng had occupied fractions of a second ago. Then Feng reached the first bounty hunter, and the man went down with his neck snapped.

Highlight (Yellow) | Location 2142

Because today was the day she’d set Mother free. Today was the day she’d touch her mommy’s mind again. Her father was going to visit the Quantum Cluster today. Going to visit her mother. And part of Ling would be there with him.

Highlight (Yellow) | Location 2155

to the entrance to the PICC – the Physically Isolated Computing Center. ***

Highlight (Yellow) | Location 2200

in this world, this world where everything was linked, where data ruled all, where cryptographic codes had replaced physical locks on the world’s wealth, on its infrastructure, on its weapons... In this world a being able to process information more rapidly than humans was the ultimate threat. It was for such reasons that the Copenhagen Accords prohibited any attempt to create a non-human self-aware being. And for the same reasons, Chen’s own government, in sponsoring the creation of exactly such a dangerous and illegal being, had taken extreme precautions that it could be physically isolated, cut off from the outside world, and destroyed remotely if necessary.

Highlight (Yellow) | Location 2367

Means, motive, and opportunity,

Highlight (Yellow) and Note | Location 2709

He'd left competitors ruined and underlings scarred in the process of making his billions. Then in later life he'd suddenly changed, become a philanthropist— a sort of midlife turnaround common among ultra-rich capitalists thinking of their legacy.

Everybody is a genius. But if you judge a fish by its ability to climb a tree, it will lead its whole life believing that it is stupid. --Einstein

Highlight (Yellow) | Location 2863

barely had time to take his seat in the opulent cabin before they were taxiing down the runway, then taking off, a pair of deadly looking fighter aircraft taking off with them. He watched the fighters out his window for a moment, before they activated their own chameleonware and became faint distortions, then nothing at all.

Highlight (Yellow) | Location 2946

This time the applause was thunderous, with hoots of approval. The sentiment line swung back hard to the right. Kim's share price cratered as Holtzmann watched, his odds of winning dropping into the gutter. In the corner of the screen, the real-time electoral map turned even more red. Holtzmann's heart sank. "Idiot." Anne clicked the screen off. "Which one?" Holtzmann asked. "Both of them."

Highlight (Yellow) | Location 2972

The Spears kidnapping in 2030. The heiress to the media fortune that included the American News Network kidnapped, dosed with DWITY, the do-what-I-tell-you drug. She'd been brainwashed, reprogrammed to siphon off part of her billions in wealth. It had been before ERD had even existed. FBI had broken the case, with Warren Becker as one of the agents. Holtzmann remembered Becker talking about it, over rounds of drinks one night, at an international Policing Emerging Technological Threats conference, in '32 or '33. Mexican cartels, Becker had said, expanding from drugs and prostitution to extortion and brainwashing.

Highlight (Yellow) | Location 3052

Saigon – Vietnam's beating heart of commerce and culture and vice. Still officially known as Hồ Chí Minh City, it was universally referred to by its older, pre-unification name.

Highlight (Yellow) | Location 3066

there was the Nexus. Half the store fronts used cheap transmitters tuned to a Nexus band, broadcasting advertisements at them. Smells and tastes came at them. The feel of fingers kneading their shoulders.

Highlight (Yellow) | Location 3075

They passed advertisements for Nexus applications that could resculpt your personality – give you confidence or adventurousness or dedication or good humor or whatever it was you felt you lacked. Be who you want to be, they said.

Highlight (Yellow) | Location 3092

In thirty days he'd need to pay again, or the tattoo would fade and his normal skin would return.

Highlight (Yellow) | Location 3106

Nexus 6 called to him. That was his real project – his way to get ahead of things, to block off most paths of abuse, to make Nexus safe again at the most basic levels of the OS. In the long term there was no way he could keep fighting individual abuses himself. There simply wasn't enough of him to keep up. He had to build something like Nexus 6 in order to solve this problem at scale.

Highlight (Yellow) | Location 3114

make their own worst fears true. Bombings and assassinations would lead to crackdowns, police and ERD retribution, witch hunts against scientists and activists, laws even worse than the Chandler Act, more loss of civil liberties in the name of "security", another step towards a police state in the USA. And that would incense the PLF, draw numbers to their ranks, drive them to ever worse atrocities, until the whole thing blew up into a full scale conflagration. Wasn't it always this way? When had terrorists ever accomplished anything but to enrage people, drive them towards greater security, greater sacrifice of freedom? They only gave their oppressors more excuses for oppression. And the oppressors just drove the oppressed further towards violent rebellion. Extremists on both sides gave power to the very forces they fought against.

Highlight (Yellow) | Location 3122

His agents had spotted the coercion code used in the attack on the President four days before the attack itself. They'd never seen the source code. Instead, they'd detected it based on a pattern of activity. Disruption of the frontal cortex to lobotomize the subject, a motor control package that turned the subject into a remote-controlled robot. It was sophisticated software, far more complex than other abuses he'd seen. Whoever had written this had invested a lot of time to write the code, even more time to test it, iterate on it until it was reliable.

Highlight (Yellow) | Location 3151

He tried to think like a PLF terrorist. His job was to program assassins. He'd want reliability, of course. That was why they were using an old version. They'd built code they knew worked. They didn't want to mess that up by upgrading to a new version and potentially introducing new bugs.

Highlight (Yellow) | Location 3424

Remote Vietnam was not a high priority target for the National Reconnaissance Office, but with more than three hundred recon birds circling in low earth orbit, now, and most of them taking frames five hundred miles square, every patch of the planet was photographed at least once an hour.

Highlight (Yellow) | Location 3453

Code that used the back doors he and Rangan had built to silently copy itself into each new mind it encountered, that hid itself from process listings and cloaked its CPU and memory usage. Libraries to rummage through the mind it entered, to alert Kade if that mind matched certain parameters, to send him back snippets of memories, contents of directories, parameters pulled from the Nexus OS.

Highlight (Yellow) | Location 3459

The OS version number was the best hook he had. Nexus OS version 0.72. If someone saw it, thought of it, had code in their mind that referred to it, Kade wanted to know.

Highlight (Yellow) | Location 3466

tapped into the white noise of a million Nexus-running minds, the surf of consciousness bathing the planet, the stuff of mind that would one day self-assemble itself into something that truly thought and felt, that could solve the problems that solitary human minds couldn't.

Highlight (Yellow) | Location 3518

Six degrees of separation, Kade thought. In days his new agent would reach every corner of the Nexus world.

Highlight (Blue) and Note | Location 3872

it. Bits of memories and conversations went through is head.

Edit

Highlight (Yellow) | Location 3882

He wanted to yell at his old self, scream at his younger self to get up, to cry out about what was going to happen, but there was no going back, no way to stop it. The past was read-only.

Highlight (Yellow) | Location 3896

So why had he missed? Nakamura's voice answered him. "To find the cause of an event... who had the most to gain?"

Highlight (Yellow) | Location 4310

Shepherd listened, then smiled and nodded herself. She turned, pointing out towards the highway, gesturing with her hands, giving directions to a lost young lady. All the while, the aerosolized DWITY variant was pumping out of Ava's blouse, making its way into Miranda Shepherd's lungs, then via her bloodstream to Shepherd's brain. Within a few seconds, Shepherd was wobbling slightly, woozy now, susceptible to suggestions. He saw Ava

reach out and take the befuddled woman's hand. The micro-injector on Ava's thumb would have just pumped more DWITY into Miranda Shepherd's bloodstream. The televangelist's wife looked confused now, her eyes glassy and blank. Ava smiled, talked soothingly, then she led Miranda Shepherd by the hand, back to the nondescript car with false plates, and drove away with her.

Highlight (Yellow) | Location 4318

As the Nigerian drove, Breece placed the voice modulator over the phone's mic, and dialed the hair salon. He spoke, and a software model trained on dozens of hours of recordings of Miranda Shepherd transformed his voice into a feminine Texas drawl.

Highlight (Yellow) | Location 4333

The net censors were back in force, tighter than he'd ever seen them. He read of the "terrorist cells" being raided and he recognized some of the names. Intellectuals. Dissidents. Those who'd dared to question the state, to propose modest reforms, or directions that differed from those the hardliners wanted.

Highlight (Yellow) | Location 4341

If the equivalence theorem existed – and it must, or why else would she have hinted at it – then any conventional computing algorithm could be incredibly accelerated via a quantum process. Quantum computing would go from a specialized technique for solving certain problems – cryptography and database searches and optimizations – to a tool that could speed up everything by billions of times, trillions of times. And he, Chen Pang, the "discoverer" of the equivalence theorem... He'd be famous, of course. A Nobelist. The greatest mind in computing since Turing. A multi-billionaire from the commercial spinoffs. One of the richest and most powerful men alive. Even in a hardliner's China, he would be untouchable, among the elite of the elite. No, he told himself. Not just for me. For the world, for the benefit to mankind. Chen nodded soberly. Yes, that was why he would do this. Not just for his own glory. But to benefit his fellow man.

Highlight (Blue) and Note | Location 4414

in London on the town in Saigon instead.

proofread

Highlight (Yellow) | Location 4940

The bullet had gone clean through her tricep, missing the bone and tendons. Her engineered clotting factors had cut off the bleeding almost immediately. Regen genes were already knitting the long fibers of her muscle back together. If Jake had had those advantages... If the technology wasn't locked up for soldiers and spies...

Highlight (Yellow) | Location 4969

She'd come here herself, almost six months ago, seeking funds. Lo Prang's illegal muay Thai fights paid well. He paid extra for an attractive Western woman willing to fight. And she, with the bookies setting the odds

steeply against her, borrowed heavily from a loan shark to bet on herself. Three fights later she'd been able to afford the new identity, the melanin therapy, the facial bone structure changes that allowed her to push on to Mae Dong.

Highlight (Yellow) | Location 4973

There was a line outside the door an hour long. Deliberate scarcity created to enforce the twin illusions of exclusivity and popularity. She went around it, towering over the women and most of the men in the line with her spiked heels. The tiny LEDs her heels were crusted with pulsed with every step, flashing as she brought them down, sending patterns of light rippling up their length. Sam headed to the front of the line, where the bouncers with their oversized muscle-grafted bodies waited, scowling, in their dark suits. She saw a look of recognition pass across one of their faces as she approached. The huge Asian man raised a finger to his ear and his lips parted as he spoke something, pitched low for a throat mic to hear. Their eyes locked, and she read those lips. Jade Tiger, they said, in Thai. Jade Tiger. His eyes stayed locked on hers, and then he nodded to the voice in his earpiece, and pulled back the velvet rope for Sam to pass.

Highlight (Yellow) | Location 5014

A sign let her know she could always tune in to ride a performer from anywhere in the world, if she liked.

Highlight (Yellow) | Location 5274

He pressed the button again. Nothing happened. He closed the controls, killed the process, relaunched it, dialed up an opiate dose again. Nothing.

Highlight (Yellow) | Location 5334

[Bruce_Lee: Block Failed] Shiva set the slate down on the writing desk, then turned and walked to the door. "I look forward to your thoughts," he said. Then he was gone, the door closed behind him. Kade sat there, reeling from the blow. When, he wondered, has that app ever fucking worked?

Highlight (Yellow) | Location 5352

He tried to imagine the consequences of the Houston bombing. Hundreds dead, maybe thousands. And the impact on US politics, on opinions of Nexus, on treatment of Nexus children... It would be the final blow. Nexus scanners would go up in schools, at bus stops, in the workplace. Security checkpoints everywhere. Maybe worse – roundups of activists, of anyone who'd protested against the Chandler Act or made pro-transhumanist statements in the past. The bomb would frighten people into accepting a security state more constrictive than ever. The PLF was playing right into the hands of the conservatives, just like terrorists always did.

Highlight (Yellow) | Location 5644

He who knows when he can fight and when he cannot will be victorious. Sun Tzu had written that in The Art of War.

Highlight (Yellow) | Location 5787

he'd never feel anyone again and no one would ever feel him and he was just empty like all the other STUPID PEOPLE who didn't have Nexus and weren't really people at all.

Highlight (Yellow) | Location 5799

They'd found a way to force Nexus out of Alfonso's brain. And now they'd do the same to Bobby. They'd cripple a little boy because it didn't fit their ideology.

Highlight (Yellow) | Location 5809

maybe if he was good and did what they wanted they'd let him go back to the other boys and still have Nexus and still be a person and still have friends and...

Highlight (Yellow) | Location 5869

What am I afraid of? he asked himself. Why don't I want to touch Shiva's thoughts? You're afraid he's telling the truth, Ilya's voice answered him. That he has only the best of intentions. Why? Because, Ilya went on. If that's the case, he has as much right to the back doors as you do. Maybe more. He's smarter than you are. He understands the world better than you do. If you deserve the back doors, then he does too. If he doesn't, then you don't either.

Highlight (Yellow) | Location 5892

Shiva took a deep breath. Am I worthy? Is this just? Is this moral? Now, as the tool he'd sought was almost in his grasp, he had his doubts. Nita would hate this, he mused. Hate it more than anything I've ever done. Hubris, she'd call it. The gods punish hubris, he told himself, in every religion, in every mythology. But he had only to think of the world outside, of the multiple precipices that humanity and this world teetered on, to hear the opposing view. That humanity needed saving. Needed it desperately. And could not do the saving itself. "I'm doing his for the world, Nita," he whispered in the darkness. "And if not me, then who? If not now, then when?"

Highlight (Yellow) | Location 5901

Kade inspected the available data. It was huge. Shiva was offering him giant swaths of thought and memory. He analyzed the files, ran them through virus checkers and security sweeps, made sure there was no embedded code in them. It was one thing to be persuaded. It was another to be tricked. He found nothing untoward. Even so, he instanced a sandbox inside his mind, and another, differently configured sandbox within the first, and only in that secure environment did he allow the files to play. He was engrossed, immediately, sucked in to what Shiva was sharing.

Highlight (Yellow) | Location 5908

Kade devoured the thoughts, the memories, the experiences, the knowledge. He bent all his Nexus CPU cycles to the task, cranked up his assimilation rate far beyond real-time. He slipped into a near trance, immersing

himself in this person, in what he knew, in what he'd done. The mask of maya slipped away, and for a time he was Kade no longer. He was Shiva, and so much more.

Highlight (Yellow) | Location 5914

He got up, went to the window, looked outside at the gorgeous water down below, the multicolored sea with its bands of jade and emerald and sea-green and lapis lazuli and a dozen more colors he couldn't name. He understood these waters, now. He knew their chemistry. He knew their ecology. He remembered diving off the coast of India, Shiva diving there, guided by his wife, examining the dying corals, despairing at their fate. Kade had read about ocean acidification. Now he understood it intimately – the horror of seeing once vibrant reefs reduced to a deathly grey. The intimate comprehension of their vulnerability, how even after Shiva's viral hack they teetered on the edge, how their death threatening all the fish and other species that depended on them.

Highlight (Yellow) | Location 5946

They lacked the cognitive capabilities necessary to collaborate on this scale. They'd done their best with democracy, with capitalism, but those had reached their limit long ago. They'd been corrupted, twisted to the interests of a few individuals, when the greatest problems the world faced were problems of collective interest. He could fix those systems.

Highlight (Yellow) and Note | Location 5954

All it would require was the key. The key that would open a million minds today, that would open tens of millions, maybe hundreds of millions of minds, at some point in the future. That was all.

until there was a competitivive open source OS, or forked copies of this, etc when people understood the backdoor manipulations. and so it goes...

Highlight (Yellow) | Location 5971

Two hundred miles off the coast of the south-eastern United States, Zoe raged. Beneath her, the October seas were hot, hotter than they'd been this late in the year in millennia. The currents of the Gulf Stream dragged warm water north from the equator and into the mid-Atlantic, adding energy to seas already heated from a record summer. The Atlantic gave off that excess heat now, evaporating it as water vapor into the air above. Zoe gorged on that warm vapor-filled air, absorbing its energy and its moisture. They added to her, strengthened her, fueling her winds, driving them ever faster and more furiously about her calm center until she whirled about at a fifth the speed of sound. North Zoe went. And chaos went with her.

Highlight (Yellow) | Location 6058

He could go into his lab, load up a syringe with a cocktail of tramadol and dapoxetine. That would do the trick. One injection, and a few minutes later, Rangan would be seizing hard,

Highlight (Yellow) | Location 7858

His heart was pounding so hard from the adrenaline, giving him strength, but shooting pain through him as well, feeling like it was about to burst. He ignored it all. Distanced himself from the pain, as the Buddhists did, stared at it dispassionately until it was just data, just sensation.

Highlight (Yellow) | Location 8197

The Science of Crux Like Nexus before it, Crux is a work of fiction, but based as accurately as possible on real science. In the extras at the back of Nexus I described the brain-implant experiments that have given humans bionic eyes and ears and the ability to control robotic arms, even from thousands of miles away. Just in the year that has passed since the writing of Nexus, more impressive work has been done. A team of researchers led by Thomas Berger has demonstrated that a digital chip can repair the impairment of a mouse's memory that occurs with brain damage to part of the brain called the hippocampus. Berger's team then went further and showed that they could improve mouse memory through the same brain implant. Another experiment by Sam Deadwyler and colleagues at Wake Forest University placed specialized brain implants in the frontal cortex of rhesus monkeys who were then trained on a 'delayed match and sample' test – a kind of monkey IQ test. Later, the monkeys had their test scores lowered by the administration of cocaine. But if the implant was switched into an active mode, it could correct this impairment, and even more. The frontal cortex device could actually raise the test scores of monkeys well beyond the scores of normal monkeys who lacked the implant. So in animals, at least, we've used brain implants to boost both memory and intelligence. Of course, the most transformative thing about the Nexus technology isn't mere augmentation – it's communication directly from one mind to another. Here also there has been progress. In an experiment by Miguel Nicolelis and colleagues, two rats, thousands of miles apart (one at Duke University in North Carolina, the other in Brazil) both had implants placed in the motor cortices of their respective brains. Nicolelis and colleagues showed that they could train one rat to respond to a series of lights by pulling the correct lever. The other rat, who had never seen these lights or levers before, would, in turn, pull the right lever most of the time, based simply on the input to its brain from the trained rat thousands of miles away. A similar study, funded by DARPA (the Defense Advanced Research Projects Agency inside the US Department of Defense), involved two monkeys, each with an implant in its auditory cortex – the part of the brain responsible for processing sound. The researchers showed that they could play a sound for one monkey and that the second monkey – in a soundproofed room – could hear that sound via the neural link, and could even identify what the sound was. The research, by the way, was conducted as part of DARPA's 'Advanced Battlefield Communications' program – a program with the goal of enhancing communication and coordination between soldiers, their squad-mates, and command. Progress towards Nexus, in short, continues apace. Crux introduces some new science, and in particular, 'uploading'. The Su-Yong Shu we see in Crux is not a flesh and blood person. Instead, she is a computer program, a vast mathematical construct of electronic neurons that initially mirrored the precise neural map of the original Su-Yong Shu's brain. For every neuron the original Su-Yong Shu had, her upload had a digital counterpart. For every synapse connecting two neurons, the upload also started with a counterpart. The idea of uploading sounds far-fetched, yet real work is happening towards it today. IBM's 'Blue Brain' project has used one of the world's most powerful supercomputers (an IBM Blue Gene/P with 147,456 CPUs) to run a simulation of 1.6 billion neurons and almost 9 trillion synapses, roughly the size of a cat brain. The simulation ran around 600 times slower than real time – that is to say, it took 600 seconds to simulate 1 second of brain activity. Even so, it's quite impressive. A human brain, of course, with its hundred billion neurons and well over a hundred trillion synapses, is far more

complex than a mouse brain. Yet computers are also speeding up rapidly, roughly by a factor 100 times every 10 years. Do the math, and it appears that a super-computer capable of simulating an entire human brain and do so as fast as a human brain should be on the market by roughly 2035 - 2040. And of course, from that point on, speedups in computing should speed up the simulation of the brain, allowing it to run faster than a biological human's. Now, it's one thing to be able to simulate a brain. It's another to actually have the exact wiring map of an individual's brain to actually simulate. How do we build such a map? Even the best non-invasive brain scanners around – a high-end functional MRI machine, for example - have a minimum resolution of around 10,000 neurons or 10 million synapses. They simply can't see detail beyond this level. And while resolution is improving, it's improving at a glacial pace. There's no indication of a being able to non-invasively image a human brain down to the individual synapse level any time in the next century (or even the next few centuries at the current pace of progress in this field). There are, however, ways to destructively image a brain at that resolution. At Harvard, my friend Kenneth Hayworth created a machine that uses a scanning electron microscope to produce an extremely high resolution map of a brain. When I last saw him, he had a poster on the wall of his lab showing a print-out of one of his brain scans. On that poster, a single neuron was magnified to the point that it was roughly two feet wide, and individual synapses connecting neurons could be clearly seen. Ken's map is sufficiently detailed that we could use it to draw a complete wiring diagram of a specific person's brain. Unfortunately, doing so is guaranteed to be fatal. The system Ken showed 'plastinates' a piece of a brain by replacing the blood with a plastic that stiffens the surrounding tissue. He then makes slices of that brain tissue that are 30 nanometers thick, or about 100,000 times thinner than a human hair. The scanning electron microscope then images these slices as pixels that are 5 nanometers on a side. But of course, what's left afterwards isn't a working brain – it's millions of incredibly thin slices of brain tissue. Ken's newest system, which he's built at the Howard Hughes Medical Institute goes even farther, using an ion beam to ablate away 5 nanometer thick layers of brain tissue at a time. That produces scans that are of fantastic resolution in all directions, but leaves behind no brain tissue to speak of. So the only way we see to 'upload' is for the flesh to die. Well, perhaps that is no great concern if, for instance, you're already dying, or if you've just died but technicians have reached your brain in time to prevent the decomposition that would destroy its structure. In any case, the uploaded brain, now alive as a piece of software, will go on, and will remember being 'you'. And unlike a flesh-and-blood brain it can be backed up, copied, sped up as faster hardware comes along, and so on. Immortality is at hand, and with it, a life of continuous upgrades. Unless, of course, the simulation isn't quite right. How detailed does a simulation of a brain need to be in order to give rise to a healthy, functional consciousness? The answer is that we don't really know. We can guess. But at almost any level we guess, we find that there's a bit more detail just below that level that might be important, or not. For instance, the IBM Blue Brain simulation uses neurons that accumulate inputs from other neurons and which then 'fire', like real neurons, to pass signals on down the line. But those neurons lack many features of actual flesh and blood neurons. They don't have real receptors that neurotransmitter molecules (the serotonin, dopamine, opiates, and so on that I talk about through the book) can dock to. Perhaps it's not important for the simulation to be that detailed. But consider: all sorts of drugs, from pain killers, to alcohol, to antidepressants, to recreational drugs work by docking (imperfectly, and differently from the body's own neurotransmitters) to those receptors. Can your simulation take an anti-depressant? Can your simulation become intoxicated from a virtual glass of wine? Does it become more awake from virtual caffeine? If not, does that give one pause? Or consider another reason to believe that individual neurons are more complex than we believe. The IBM Blue Gene neurons are fairly simple in their mathematical function. They take in inputs and produce outputs. But an amoeba, which is both smaller and less complex than a human neuron, can do far more. Amoebae hunt. Amoebae remember the places they've found food. Amoebae choose which direction to propel themselves with their flagella. All of those

suggest that amoebae do far more information processing than the simulated neurons used in current research. If a single celled micro-organism is more complex than our simulations of neurons, that makes me suspect that our simulations aren't yet right. Or, finally, consider three more discoveries we've made in recent years about how the brain works, none of which are included in current brain simulations. First, there're glial cells. Glial cells outnumber neurons in the human brain. And traditionally we've thought of them as 'support' cells that just help keep neurons running. But new research has shown that they're also important for cognition. Yet the Blue Gene simulation contains none. Second, very recent work has shown that, sometimes, neurons that don't have any synapses connecting them can actually communicate. The electrical activity of one neuron can cause a nearby neuron to fire (or not fire) just by affecting an electric field, and without any release of neurotransmitters between them. This too is not included in the Blue Brain model. Third, and finally, other research has shown that the overall electrical activity of the brain also affects the firing behavior of individual neurons by changing the brain's electrical field. Again, this isn't included in any brain models today. I'm not trying to knock down the idea of uploading human brains here. I fully believe that uploading is possible. And it's quite possible that every one of the problems I've raised will turn out to be unimportant. We can simulate bridges and cars and buildings quite accurately without simulating every single molecule inside them. The same may be true of the brain. Even so, we're unlikely to know that for certain until we try. And it's quite likely that early uploads, like Su-Yong Shu, will be missing some key piece or have some other inaccuracy in their simulation that will cause them to behave not-quite-right. Perhaps it'll manifest as a creeping insanity, as in Su-Yong's case. Perhaps it will be too subtle to notice. Or perhaps it will show up in some other way entirely. Finally, I've written about more than neuroscience in Crux. And in particular I've written about the impact of climate change. Zoe, the storm that hits the eastern seaboard at the end of Crux, is a piece of fiction, but a plausible one. When I wrote the scenes with Zoe, in late 2012, superstorm Sandy had not yet appeared. (Imagine my surprise, when, a few weeks after I wrote about it, a late season storm struck the eastern seaboard and impacted a presidential election!) Since then, most of the public has learned that hurricanes can indeed arrive in early November, and that they feed off the power of warm surface waters in the Atlantic. It's impossible to say that a changing climate caused a particular storm. But what it is possible to say is that the general warming we've experienced has made storms like Zoe (and Sandy) many times more likely to occur. As the planet continues to warm, we'll see far more of them. Similarly, the crime for which Shiva was exiled, the deliberate release of a virus that genetically tweaked coral reefs to increase their odds of survival, is not so completely far-fetched. As oceans warm and grow more acidic, corals have a harder and harder time surviving. The best estimate is that, by 2100, roughly half of all coral species in existence today will go extinct. But, of course, there are some corals that thrive in extremely warm water or extremely acidic water. They're able to because of various genes they carry that have evolved for those conditions. In 2012, two research projects set out to find the genes that allow some coral species to survive in warm waters. It's not a large leap to imagine a similar project searching for acidity-survival genes. And from there, the idea of deliberately transplanting such genes into corals that are struggling becomes rather tempting. Of course, this would certainly constitute the release of a 'GMO' into the wild, and it would be met with tremendous opposition. (A simpler, less controversial, path would be to transport coral 'cuttings' from hardy species into areas where corals are dying. Of course, given the millions of square miles of coral reefs in the world's oceans, that would also be incredibly more labor intensive.) If you're interested in more on human enhancement and particularly the frontiers of neuroscience, you may want to read my non-fiction book, *More Than Human: Embracing the Promise of Biological Enhancement*. If you're interested in the impact of climate change and other natural resource challenges – and the science and technology to overcome those problems – you may like my other non-fiction book, *The Infinite Resource: The Power of Ideas on a Finite Planet*. And of course, if you enjoyed this book, the very nicest thing you can do for me is to let the world know, by telling your

friends, by posting about it online, and by leaving a review of this book at Amazon or whatever sites you use to buy and discover books.
